

# HW 1 - Multiple Linear Regression

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm
```

```
In [2]: # Import raw data file

input_file = "On_Time_Reporting_Carrier_On_Time_Performance_(1987_present)_2026_1.csv"
df = pd.read_csv(input_file)

condition = (df["Reporting_Airline"] == "UA") & (df["Origin"] == "ORD")
# condition = df["Origin"] == "ORD"

df_filtered = df[condition]

columns_to_keep = [
    # "DayofMonth",
    "DayOfWeek",
    # "Reporting_Airline",
    # "Origin",
    # "Dest",
    "DepTime",
    "TaxiOut"
]

fdata = df_filtered[columns_to_keep].copy()

for col in columns_to_keep:
    fdata[col] = pd.to_numeric(fdata[col], errors="coerce")

fdata = fdata.dropna(subset=columns_to_keep) # delete N/A data

hours = fdata["DepTime"] // 100 # change "hhmm" to [min] of the day
minutes = fdata["DepTime"] % 100

fdata["DepTime_Min"] = np.where(
    fdata["DepTime"] > 0, hours * 60 + minutes, 0.0
)
```

```
# fdata.to_csv("dataset_cleaned.csv", index=False)
print(fdata.head())
```

```
/var/folders/vc/nml18my11hs68jjym7g_y_c40000gn/T/ipykernel_38871/1531807560.py:4: DtypeWarning: Columns (0: Div2Airport, 1: Div2TailNum) have mixed types. Specify dtype option on import or set low_memory=False.
```

```
df = pd.read_csv(input_file)
```

|        | DayOfWeek | DepTime | TaxiOut | DepTime_Min |
|--------|-----------|---------|---------|-------------|
| 480223 | 6         | 1800.0  | 19.0    | 1080.0      |
| 480238 | 6         | 741.0   | 28.0    | 461.0       |
| 480240 | 6         | 1622.0  | 48.0    | 982.0       |
| 480256 | 6         | 1102.0  | 28.0    | 662.0       |
| 480258 | 6         | 1259.0  | 22.0    | 779.0       |

### Dataset Explanation

The primary objective of this analysis is to investigate the operational factors influencing airplane taxi-out time. I hypothesize that taxi-out duration is predominately driven by temporal demand patterns, specifically the day of the week (DayOfWeek) and the actual departure time of day (DepTime). To encode the categorical variable DayOfWeek, I utilized dummy coding with 7-Sun set as the reference baseline.

To eliminate other variations, I focused on a controlled subset: United Airlines (UA) flights departing from Chicago O'Hare International Airport (ORD).

In addition, the day of week dummy variables are binary, whereas DepTime\_Min ranges up to 1440 minutes. Therefore, the following predictor matrix is normalized to [-1, 1] to improve numerical conditioning.

```
In [3]: # Scatter Plot

plot_vars = ["DayOfWeek", "DepTime_Min", "TaxiOut"]

axes = pd.plotting.scatter_matrix(
    fdata[plot_vars],
    marker="o",
    s=5,
    alpha=0.3,
    diagonal='kde'
)

tick_mins = [0, 360, 720, 1080, 1440]
tick_labels = ["00:00", "06:00", "12:00", "18:00", "24:00"]

for r in range(3):
```

```

axes[r, 1].set_xlim([0, 1440])
axes[r, 1].set_xticks(tick_mins)

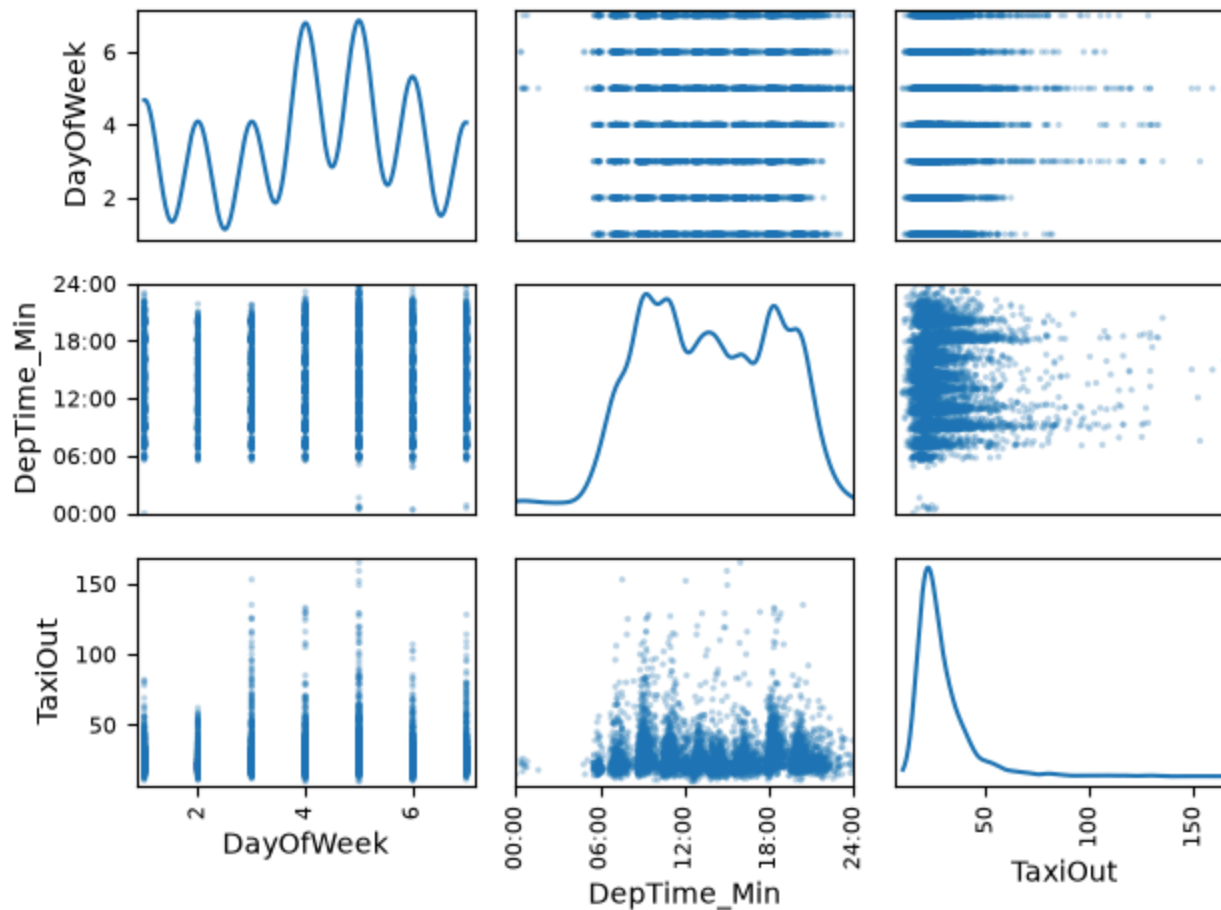
axes[2, 1].set_xticklabels(tick_labels)

for c in range(3):
    if c != 1:
        axes[1, c].set_ylim([0, 1440])
        axes[1, c].set_yticks(tick_mins)

axes[1, 0].set_yticklabels(tick_labels)

plt.tight_layout()
plt.show()

```



## Scatter Plot Explanation

This scatter plot shows the relationships among three variables: DayOfWeek, DepTime\_Min, and TaxiOut.

- Diagonal Plots
  - DayOfWeek: Flight operations are distributed across all days of the week; higher traffic volumes concentrated toward Thr, Fri, and Sat.
  - DepTime\_Min: Departures are concentrated between 06:00 and 22:00; showing two distinct operational peaks—one in the late morning and another in the late afternoon.
  - TaxiOut: The distribution is right-skewed; while the majority of data cluster around 30 minutes, notable outliers extend beyond normal operational ranges.
- Off-Diagonal Plots
  - DayOfWeek vs. DepTime\_Min: Flight departure schedules are uniformly spread across the operational hours regardless of the specific day of the week.
  - DayOfWeek vs. TaxiOut: Most taxi-out durations remain generally uniform across the week, but extreme long-tail delays occur prominently on Wed and Fri.
  - DepTime\_Min vs. TaxiOut: Taxi-out durations are lowest during early morning and late night hours; sporadically large taxi-out delays appear throughout the daytime operational window.

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In [4]: *# X normalized function*

```
def normalize_data(X):
    """Function to normalize input data to [-1, 1].

    Parameters
    -----
    X : array_like
        Input data (n x (p + 1) dimensions), where each row is an observation and each column is a predictor. The first

    Returns
    -----
    X_normalized : array_like
        Input data (n x (p + 1) dimensions) normalized to the range [-1, 1].

    """

    n, n_columns = np.shape(X)

    # normalize each column
    X_normalized = np.ones((n, n_columns))
```

```

for column in range(n_columns):
    x_column = X[:, column]
    x_max = max(x_column)
    x_min = min(x_column)
    if x_max != x_min:
        x_column_normalized = [
            (x - (x_max + x_min) / 2) / ((x_max - x_min) / 2) for x in x_column
        ]
        X_normalized[:, column] = x_column_normalized

return X_normalized

```

In [5]: *# Make X list: [1, "Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "DepTime\_Min"]*

```

for day_code in range(1,7):
    col_name = f"Day_{day_code}"
    fdata[col_name] = (fdata["DayOfWeek"] == day_code).astype(int) # delete 7-Sun

day_dummy_cols = [f"Day_{i}" for i in range(1,7)]
X_cols = day_dummy_cols + ["DepTime_Min"]

X_temp = fdata[X_cols].values
n = len(fdata) # num of data

X = np.concatenate((np.ones((n, 1)), X_temp), axis=1) # put beta_0 column
X_normalized = normalize_data(X) # normalized

y = fdata["TaxiOut"].values.reshape(-1, 1)

```

In [6]: *# Utilizing `statsmodels` library*

```

results = sm.OLS(y, X_normalized).fit()

X_cols_name = ["const."] + X_cols

# (1) ANOVA model test, (2) individual coefficient hypothesis tests
print(results.summary(xname=X_cols_name))

```

## OLS Regression Results

```

=====
Dep. Variable:          y      R-squared:          0.030
Model:                  OLS    Adj. R-squared:       0.029
Method:                 Least Squares  F-statistic:      31.63
Date:                  Sun, 13 Sep 2026  Prob (F-statistic):  1.71e-43
Time:                  22:51:18  Log-Likelihood:    -28505.
No. Observations:      7067    AIC:              5.703e+04
Df Residuals:          7059    BIC:              5.708e+04
Df Model:              7
Covariance Type:       nonrobust
=====

```

|             | coef    | std err | t      | P> t  | [0.025 | 0.975] |
|-------------|---------|---------|--------|-------|--------|--------|
| const.      | 26.4671 | 1.058   | 25.007 | 0.000 | 24.392 | 28.542 |
| Day_1       | -1.3596 | 0.318   | -4.282 | 0.000 | -1.982 | -0.737 |
| Day_2       | -2.1055 | 0.324   | -6.494 | 0.000 | -2.741 | -1.470 |
| Day_3       | 0.4415  | 0.324   | 1.361  | 0.174 | -0.194 | 1.078  |
| Day_4       | -0.4195 | 0.302   | -1.389 | 0.165 | -1.012 | 0.173  |
| Day_5       | 1.7087  | 0.302   | 5.662  | 0.000 | 1.117  | 2.300  |
| Day_6       | -1.1703 | 0.313   | -3.742 | 0.000 | -1.783 | -0.557 |
| DepTime_Min | 0.6359  | 0.431   | 1.476  | 0.140 | -0.209 | 1.481  |

```

=====
Omnibus:                 4929.252    Durbin-Watson:          1.859
Prob(Omnibus):            0.000    Jarque-Bera (JB):        89103.844
Skew:                     3.153    Prob(JB):                 0.00
Kurtosis:                 19.212    Cond. No.                 15.1
=====

```

### Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```

In [7]: print(f"F-statistic: {results.fvalue:.8f}")
        print(f"p-value: {results.f_pvalue:.10e}")

```

```

F-statistic: 31.62521101
p-value: 1.7084670362e-43

```

### (1) ANOVA model test

The model yielded an F-statistics of 31.63 with DoF of model 7 and DoF of residuals 7059, corresponding to a p-value of 1.71e-43. Because the p-value is substantially lower than the standard threshold, we reject the null hypothesis and confirm that at least one of the predictors in this model is statistically associated with taxi-out duration.

```
In [8]: for name, coef, tval, pval in zip(X_cols_name, results.params, results.tvalues, results.pvalues):
        print(f"{name:<12}: coef = {coef:10.6f}, t-value = {tval:10.4f}, p-value = {pval:10.8e}")
```

```
const.      : coef = 26.467149, t-value = 25.0069, p-value = 2.57770325e-132
Day_1       : coef = -1.359585, t-value = -4.2816, p-value = 1.88001629e-05
Day_2       : coef = -2.105505, t-value = -6.4944, p-value = 8.90536670e-11
Day_3       : coef = 0.441532, t-value = 1.3609, p-value = 1.73589016e-01
Day_4       : coef = -0.419547, t-value = -1.3886, p-value = 1.64987695e-01
Day_5       : coef = 1.708663, t-value = 5.6622, p-value = 1.55288521e-08
Day_6       : coef = -1.170311, t-value = -3.7419, p-value = 1.84092445e-04
DepTime_Min : coef = 0.635925, t-value = 1.4756, p-value = 1.40105350e-01
```

## (2) Individual coefficient hypothesis tests

- DepTime\_Min: This is not statistically significant at the 5% significance level because  $p = 0.140$ . Although its estimated coefficient is positive, there is insufficient evidence to conclude that DepTime\_Min significantly affects taxi-out duration after accounting for day of the week.
- Day\_5: Fri exhibits statistically significant positive coefficient, indicating a significantly longer taxi-out duration compared with Sun.
- Day\_1, Day\_2 and Day\_6: Mon, Tue and Sat yield statistically significant negative coefficients, demonstrating substantially less ground queuing than Sunday.
- Day\_3 and Day\_4: Wed and Thu fail to reject the null hypothesis at the 5% significance level, showing less statistically discernable difference in taxi-out duration relative to Sun.

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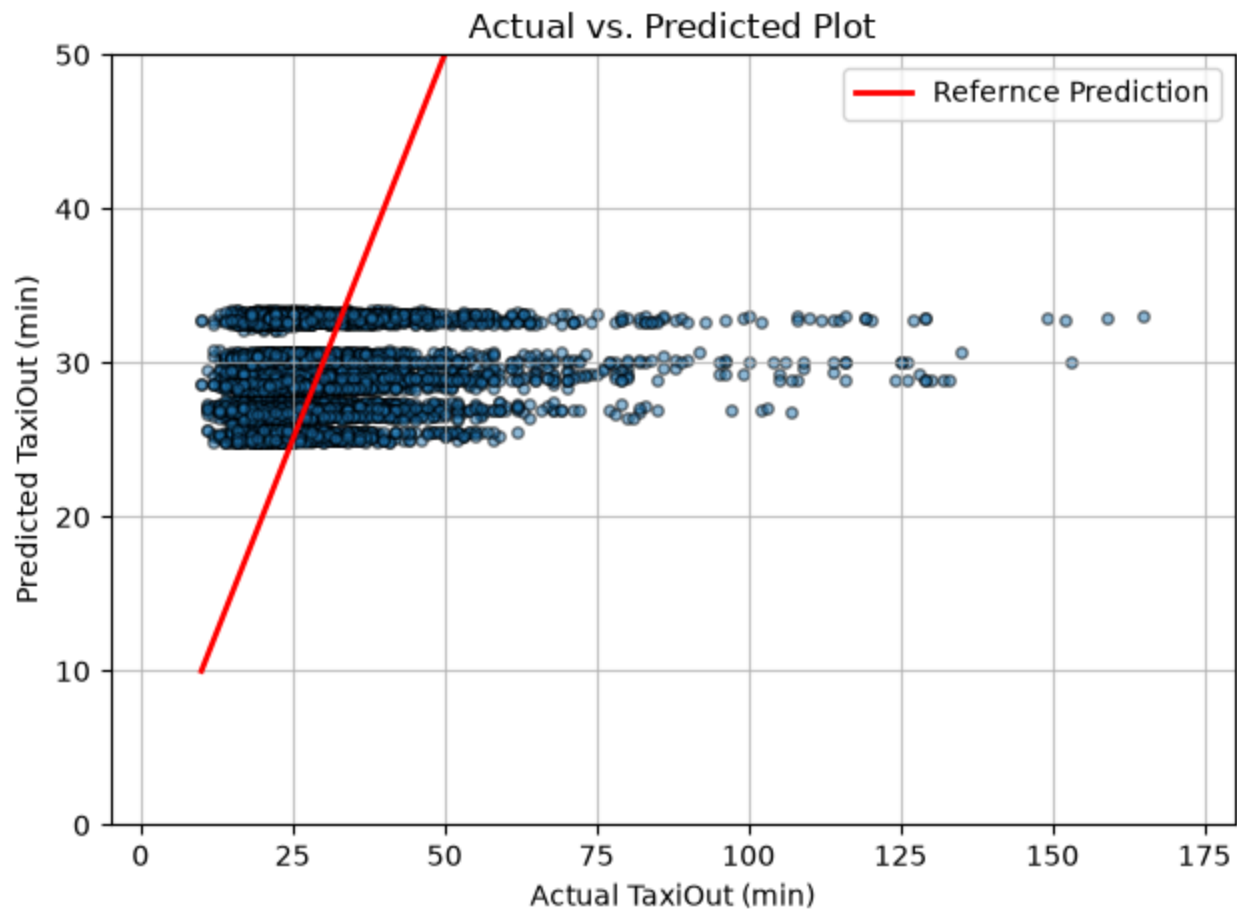
```
In [9]: # (3) Actual vs. Predicted plot
```

```
y_pred = results.predict(X_normalized)
y_actual = y.flatten()

plt.figure()
plt.scatter(y_actual, y_pred, alpha=0.5, edgecolors='k', s=15)

min_val = min(y_actual.min(), y_pred.min())
max_val = max(y_actual.max(), y_pred.max())
plt.plot(
    [min_val, max_val],
    [min_val, max_val],
    color="red",
    linewidth=2,
    label="Reference Prediction",
)
```

```
plt.title("Actual vs. Predicted Plot")
plt.xlabel("Actual TaxiOut (min)")
plt.ylabel("Predicted TaxiOut (min)")
plt.xlim([-5, 180])
plt.ylim([0, 50])
plt.grid(True, alpha=0.7)
plt.legend(frameon=True)
plt.tight_layout()
plt.show()
```



### (3) Actual vs. Predicted plot

The red line represents the ideal reference prediction; however, the data points do not follow this line, clustering horizontally within a narrow band of 25-35 minutes and failing to capture extreme tail delays.



This behavior is consistent with the model's low coefficient of  $R\_squared = 0.030$ , indicating that the current feature set explains only a small fraction of the total variance in taxi-out duration. Although the overall regression model is statistically significant, the current predictors are insufficient on their own to model the real nature of airport ground operations.

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